

Complements of Operational Research

6.1 Queueing: concepts, Little's law and M/M/1

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Why study queues?

Queues form whenever demand arrives **irregularly** at a service with limited capacity: hospital emergency departments, call centres, toll plazas, supermarket checkouts, electric-vehicle chargers, web servers.

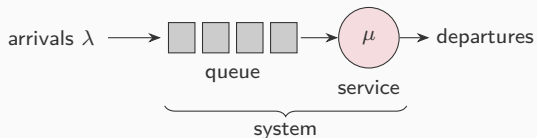
The trade-off

More capacity (servers, people) costs money; less capacity makes customers wait — and waiting also costs (lost customers, idle machines, dissatisfaction).

A surprise: even with capacity *above* the mean demand there is waiting — and it can be long. The culprit is **variability**: sometimes several arrive in a row, sometimes a service takes longer.

Queueing theory (Erlang, 1909, Copenhagen telephone exchanges) gives formulas for the steady state: on average, how many wait and for how long.

Elements and notation



- λ : mean arrival rate (customers per unit of time);
- μ : mean service rate of *one* server;
- s : number of servers; $\rho = \lambda/(s\mu)$: utilisation.

Kendall notation: $A/B/s/\text{discipline}/K/N$

- A, B : arrivals and service; **M** = exponential (Markov);
- s servers; K = maximum capacity; N = population;
- discipline: FIFO, LIFO, SIRO or GD (general).

By default $K = \infty$, $N = \infty$, FIFO: $M/M/1 = M/M/1/GD/\infty/\infty$.

Measures: L , L_q (mean number in system / in queue), W , W_q (mean time in system / in queue), P_n .

Little's law (1961)

In steady state,

$$L = \lambda W, \quad L_q = \lambda W_q, \quad W = W_q + \frac{1}{\mu}.$$

It holds for **any** stable queue (any distribution, any discipline): knowing one of the four measures is enough to get the others.

Intuition: if 3 cars arrive per hour and each one stays, on average, 1 hour, then at any instant there are, on average, $3 \times 1 = 3$ cars there.

Practical use: in a shop with 40 customers inside, on average, and 120 entries per hour, each customer stays on average $40/120 \text{ h} = 20 \text{ min}$ — without timing anyone.

The example: a fast-charging station

A service station has **one** fast charger. On average, $\lambda = 3$ cars arrive per hour (Poisson arrivals); a charge takes, on average, 15 min ($\mu = 4$ per hour, exponential).

M/M/1 ($\rho = \lambda/\mu < 1$)

$$P_0 = 1 - \rho, \quad P_n = (1 - \rho)\rho^n,$$

$$L = \frac{\rho}{1 - \rho}, \quad L_q = \frac{\rho^2}{1 - \rho},$$

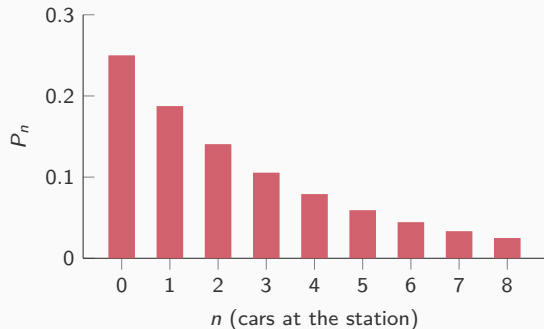
$$W = \frac{1}{\mu - \lambda}, \quad W_q = \frac{\lambda}{\mu(\mu - \lambda)}.$$

Measure	Value	Meaning
ρ	0.75	busy 75% of the time
P_0	0.25	free 25% of the time
L	3	cars at the station
L_q	2.25	cars waiting
W	1 h	waiting + charging
W_q	45 min	waiting only

Little confirms: $L = \lambda W = 3 \times 1 = 3$ and $L_q = 3 \times 0.75 = 2.25$.

A 15-min charge costs, on average, 45 min of waiting.

How many cars are at the station?

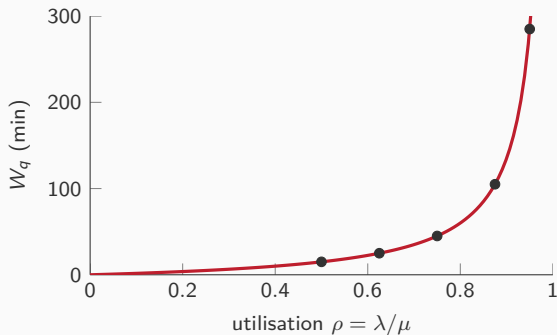


$P_n = (1 - \rho)\rho^n$ decreases geometrically.

- $P(\text{wait}) = P(n \geq 1) = \rho = 75\%$;
- $P(n \geq 3) = \rho^3 = 42\%$: almost half of the time there are at least 2 cars waiting;
- $P(n \geq 8) = \rho^8 = 10\%$.

The mean ($L = 3$) hides a lot of variation: there are quiet hours and hours with a long queue.

The effect of utilisation: waiting explodes near 100%



λ (/h)	ρ	W_q
2	0.50	15 min
2.5	0.625	25 min
3	0.75	45 min
3.5	0.875	1 h 45
3.8	0.95	4 h 45

17% more demand ($3 \rightarrow 3.5$ cars/h) and the waiting time **more than doubles**.

That is why a service is not sized for 95% occupancy: W_q grows like $\rho/(1 - \rho)$.

Confirming with simulation — and seeing the tail

```
import numpy as np
rng = np.random.default_rng(1)
lam, mu, n = 3, 4, 200_000
chegadas = np.cumsum(rng.exponential(1/lam, n))
servico = rng.exponential(1/mu, n)
espera = np.zeros(n); fim = 0.0
for i in range(n):
    inicio = max(chegadas[i], fim)      # wait for the previous car
    espera[i] = inicio - chegadas[i]
    fim = inicio + servico[i]
e = espera[10_000:]                    # discard the warm-up
print(e.mean()*60, (e > 0).mean(), np.quantile(e, .95))
# 45.5 min    0.75    2.72 h
```

	formula	simulation
W_q	45 min	45.5 min
$P(\text{wait})$	0.75	0.75
95% wait less than	2.71 h	2.72 h

The mean wait is 45 min, but **1 in every 20 drivers waits more than 2 h 40**. In the M/M/1, $P(W_q > t) = \rho e^{-(\mu-\lambda)t}$.

Simulation does not need the assumptions (for example, charges of almost fixed duration) — the formulas do.

Example 1 — baggage screening

At Porto Santo airport there is a single X-ray machine to check baggage. The airport receives, on average, 10 passengers per minute (exponential interarrival times) and the screening serves 12 people per minute (exponential).

- a) What is the probability that the X-ray machine is not idle?
- b) What is the probability that there are passengers in the queue?
- c) On average, how many passengers are in the queue?
- d) On average, how much time does a passenger need to get through the security area?

KEY TAKEAWAYS — 6.1

1. $\lambda, \mu, \rho = \lambda/(s\mu)$; in infinite-capacity models, stability requires $\rho < 1$. Kendall notation; measures L, L_q, W, W_q, P_n .
2. Little's law: $L = \lambda W, L_q = \lambda W_q, W = W_q + 1/\mu$ — for any stable queue.
3. M/M/1: $L = \rho/(1 - \rho), W = 1/(\mu - \lambda), P_n = (1 - \rho)\rho^n$.
4. Waiting grows like $\rho/(1 - \rho)$: near 100% utilisation, small increases in demand give long waits.
5. The mean hides the tail: simulate (or use $P(W_q > t)$) to know how many wait a long time.

Next: 6.2 Limited capacity and several servers

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Queueing Theory.
- Gross, D., Shortle, J. F., Thompson, J. M., Harris, C. M., *Fundamentals of Queueing Theory*, 4th ed., Wiley, 2008.
- Little, J. D. C., A proof for the queuing formula $L = \lambda W$, *Operations Research*, 9(3), 383–387, 1961.

Resources

- Exercise sheet 6. `code/python/06_queueing/06_1_mm1/filas.py` (formulas and simulation).